AY 26/27

CM3191 - Chemical Synthesis Experiments

ORGANIC LAB MANUAL

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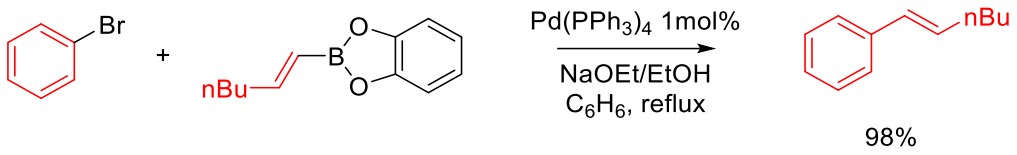
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Experiment 1: Suzuki-Miyaura Cross-Coupling in Water

I. Introduction

The Suzuki-Miyaura cross-coupling reaction, named after its pioneers Akira Suzuki and Norio Miyaura, has revolutionized the field of organic synthesis. This palladium-catalyzed reaction enables the formation of carbon-carbon bonds by coupling aryl or vinyl boronic acids with aryl or vinyl halides or triflates.



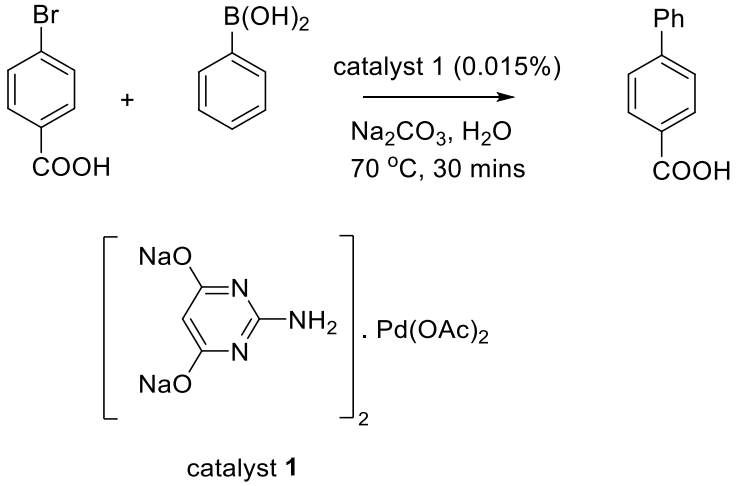
Suzuki and Miyaura 1979

The Suzuki-Miyaura cross-coupling produces biaryl and vinyl-aryl compounds. They are important targets or intermediates in various sectors, particularly the pharmaceutical industry. Examples of pharmaceutical products that used Suzuki cross-coupling in their syntheses are abemaciclib, an anti-cancer drug that inhibits cell cycle progression; rucaparib, an FDA-approved PARP inhibitor used in ovarian cancer treatment; lapatinib, used to treat HER2-positive breast cancer and many more.

The growing emphasis on sustainable and environmentally friendly processes has led to the adaptation of green chemistry principles in many organic reactions, including the Suzuki-Miyaura reaction. Green chemistry aims to minimize the environmental impact of chemical processes by reducing the use of hazardous substances, increasing energy efficiency, and enhancing the overall safety and sustainability of chemical manufacturing.

In this experiment, you will explore an approach to make organic reactions greener: the use of renewable and greener solvents. Traditional Suzuki reactions often employ organic solvents like toluene and DMF, which are harmful to the environment. Green chemistry encourages the use of renewable and less toxic solvents such as water, ethanol, or solvent-free conditions. When you do this experiment, think about the design: what are the challenges of carrying out an organic reaction in water? And why does this reaction work well in this solvent?

II. Reaction Scheme



catalyst **1**

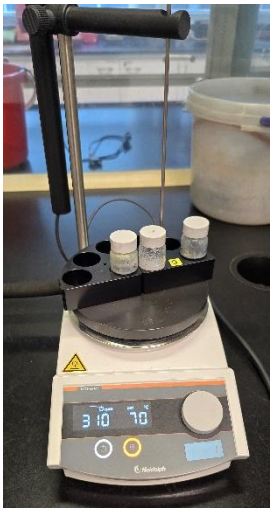
Reagents and Product Table

Reagent	MW or conc	mmol	amount
4-bromobenzoic acid	201.02	1.7	0.34 g
phenylboronic acid	121.93	____	0.25 g
Pd catalyst solution	0.25 mM	____	1.0 mL
Na_2CO_3 solution	0.5 M	____	10 mL
Product	198.22	____	____

Safety Precaution: *4-Bromobenzoic acid, phenylboronic acid, sodium carbonate, 2-amino-4,6-dihydroxyaminopyrimidine, and palladium acetate may be harmful if inhaled, swallowed, or absorbed through the skin. Palladium acetate and sodium carbonate are irritants. The catalyst solution should be considered harmful. Sodium hydroxide is caustic and can cause burns to skin, eyes and respiratory tract. Hydrochloric acid is corrosive and can cause burns to skin, eyes, and respiratory tract. Ethanol is highly flammable and can cause skin and eye irritation. DMSO-d6 is absorbed readily through the skin and may transport harmful substances into the body. The product of the coupling, 4-phenylbenzoic acid, is not considered harmful, but care should be taken to avoid inhalation or contact with skin. Do not inhale chemicals or allow them to contact the skin and/or eyes. Perform all operations in a well-ventilated fume cupboard. Refer to the Risk Assessment for detailed safety information.*

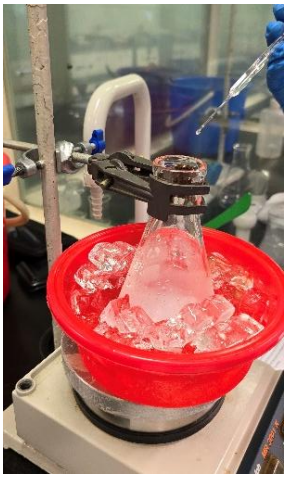
III. Procedure:[1]

3.1. Reaction setup



- Turn on the hotplate stirrer equipped with heating blocks and set the temperature to 70 °C (Note: this is a shared equipment).
- Add 0.34 g of 4-bromobenzoic acid, 0.25 g of phenylboronic acid and a stir bar to a 20 mL reaction vial.
- Add 10 mL Na_2CO_3 solution (0.5 M) to the reaction vial, close the cap and stir the resulting mixture vigorously, using the hotplate stirrer in your assigned fumehood (not the common hotplate at 70 °C), until all the reactants are dissolved.
- Add 1 mL of the palladium catalyst solution (0.25 mM) to the reaction vial, close the cap and place the vial in the preheated heating block at 70 °C. Allow the reaction to run for 30 minutes.

3.2. Reaction workup





- Take the reaction mixture off the hotplate and cool it to room temperature. Transfer the mixture into a 250 mL conical flask and cool the conical flask in an ice bath, with stirring.
- While the mixture is stirring, slowly add 20 mL of 1M HCl into the flask dropwise (note: gas will form). Check that the mixture is acidic with pH paper. Add 20 mL of water and continue to stir the mixture for 10 minutes in the ice bath.
- Isolate the crude product using suction filtration.

[1] *Adapted from Alaina E. Hamilton, Audrey M. Buxton, Christopher J. Peeples, and Justin M. Chalker, An Operationally Simple Aqueous Suzuki–Miyaura Cross-Coupling Reaction for an Undergraduate Organic Chemistry Laboratory. J. Chem. Educ. 2013, 90, 1509–1513.*

3.3. Purification: recrystallization with single solvent: ethanol

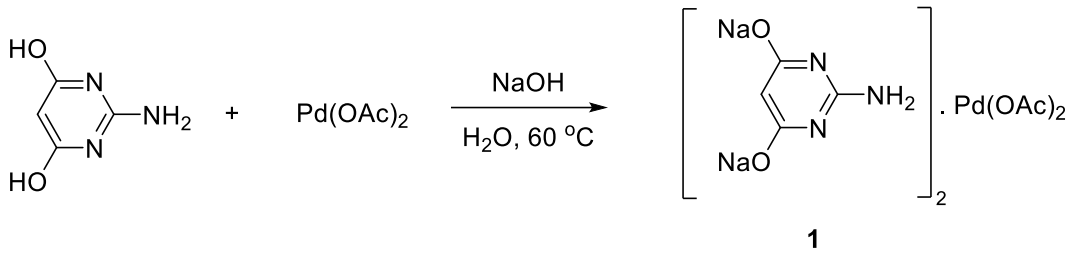


- Transfer the crude product into a 50 mL conical flask (or a 50 mL beaker) with a stir bar. Add 3 mL of 1M HCl and 10 mL of ethanol to the mixture (solution 1).
- In another 50 mL conical flask, add 20 mL of ethanol and a few boiling chips (solution 2).
- Heat the two solutions to gentle boiling. Using a dropper, slowly add just enough warm ethanol from solution 2 to solution 1 to dissolve all the material. Note: you should use a minimal amount of ethanol to dissolve the solid.
- Cool the solution to room temperature, undisturbed (about 10-15 mins). Then, place both solutions 1 and 2 in an ice bath for 15 minutes to complete the recrystallisation.
- Isolate the purified product using suction filtration. Use a minimal amount of cold ethanol (solution 2) to transfer and wash the product, if necessary.
- Transfer the crystals to a large filter paper labelled with your name, and leave it to dry under an IR lamp for 15 mins.

3.4. Characterization

- Transfer the product into a pre-weighed plastic bag. Weigh and record the mass of the purified product. Calculate the percentage yield.
- Obtain an IR spectrum of the product.
- Prepare a sample for NMR analysis: weigh about 10 mg of the sample in a clean and dry plastic vial. Dissolve it in 0.6 mL of the NMR solvent and transfer the solution into an NMR tube using a clean and dry glass dropper. Ensure the height of the solution in the NMR tube is about 5 cm and top it up with the same NMR solvent if needed. Cap and label the NMR tube.

Appendix 1: Preparation of palladium catalyst *(by staff)*



1

Add palladium acetate (5.6 mg, 0.025 mmol) and 2-amino-4,6-dihydroxypyrimidine (6.3 mg, 0.050 mmol) to a 50 mL beaker. In a separate beaker, add sodium hydroxide (4.0 mg, 0.10 mmol) to 10 mL of deionised water and stir until dissolved. Transfer the solution of sodium hydroxide into the first beaker containing the palladium acetate and the ligand. Heat the stirred mixture in a 60°C water bath for 20 minutes or until all materials have dissolved. The solution should appear yellow-orange and have no solid material. Finally, transfer the catalyst solution into a 100 mL volumetric flask and fill to line with deionised water. This concentration of palladium in this catalyst solution is 0.25 mM and should appear pale yellow.

Name: _____

Date of the experiment: _____

Datasheet for Suzuki-Miyaura Cross-Coupling in Water

Instruction: Take photos of the measurements. After completing the experiment, show the photos to your teaching assistant and ask them to sign your printed datasheet. Submit the signed datasheet along with your lab report

Starting materials

- Actual mass of 4-bromobenzoic acid used (g)_____
- Actual mass of phenylboronic acid used (g)_____

Product

- Mass of empty plastic bag (g)_____
- Mass of plastic bag + product (g) _____
- Mass of your final product (g) _____
- **Percentage yield (%):** _____

Observation during the experiment:

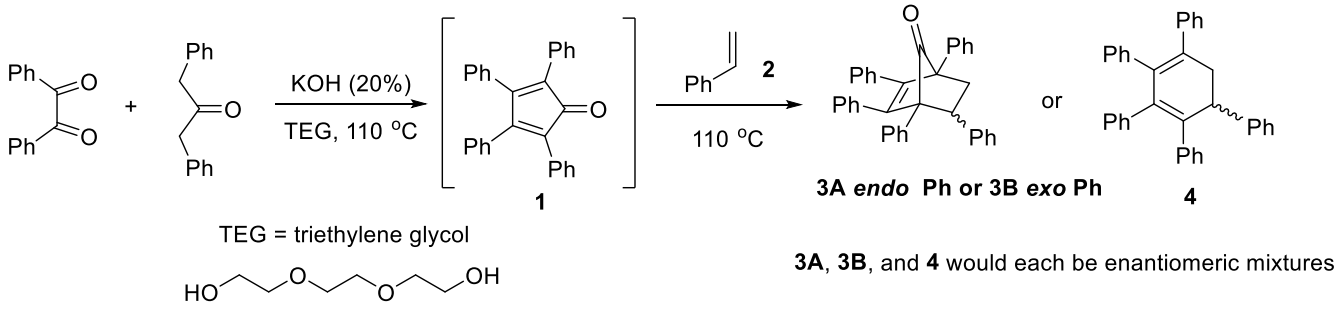
Experiment 2: One-pot Aldol Condensation and Diels-Alder Reaction

I. Introduction

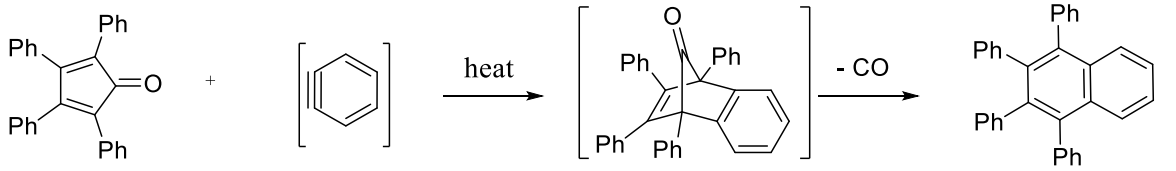
The aldol condensation and Diels-Alder reactions are two important C-C bond forming transformations exploited by synthetic chemists. They are also two reactions often cited as representing green chemical processes because of their high atom economy and flexible solvent requirements. In this experiment these two reactions will be combined in a one pot process which saves the energy and solvents required to workup and isolate the intermediate compound.

In the first step, 1,3-diphenylacetone and benzil are reacted with ethanolic KOH in triethylene glycol solvent. This multi-step aldol condensation yields the highly conjugated tetraphenylcyclopentadienone **1** (often called tetracyclone in the old literature). This uniquely colored (deep purple/black) compound can then serve as the diene in a Diels-Alder reaction. Treatment with styrene **2**, as the dienophile, and heating yields a white solid which will be recrystallized and characterized by melting point, IR and ^{1}H NMR spectroscopy. Possible products include the diastereomeric *endo* and *exo* bridged compounds, **(3A)** and **(3B)**, which upon formation could also lose carbon monoxide to yield the cyclohexadiene compound, **(4)**; each product would be expected to be a racemic mixture of enantiomers.

Reaction scheme



Precedence for the decarbonylation is seen in the reaction of tetraphenylcyclopentadienone with benzyne, illustrated below.



only product isolated

With respect to ^{1}H NMR spectroscopy, all three of the proposed compounds (**3A**, **3B**, and **4**) have three non-aromatic hydrogens which can be studied. The CH_2 protons are diastereotopic because of the neighboring stereocenter (chiral carbon). Therefore, each compound is expected to have three doublets of doublets whose coupling constant values are related to their bond angles. The geminal (on the same carbon) hydrogens should have a large coupling constant on the order of 11-14 Hz. The coupling constants of hydrogens on adjacent carbons will be dependent upon their dihedral angle and can be predicted by the Karplus relationship. Note, that in the bicyclic structures, **3A** and **3B**, the ring is locked in a fixed geometry with eclipsing hydrogens, in contrast to axial and equatorial positions observed for the flexible cyclohexane ring.

Though study of the IR spectrum and ^{1} H NMR chemical shifts and coupling constants can infer which structure was isolated, the precise stereochemistry can be unequivocally observed in the X-ray crystal structure. The solved structure of the product will be provided for you to examine the structure of the product.

II. Reagents and Product Table

Reagent	MW or conc	mmol	Density (g/mL)	amount
Benzil	210.23	____	NA	0.21 g
1,3-diphenylacetone	210.27	____	NA	0.21 g
triethylene glycol	150.17	NA	1.113	0.50 mL
KOH 2M	56.10	cat	NA	5-6 drops
Styrene	104.15	____	0.906	0.40 mL
Product 3A or B	488.63	____	NA	___
Product 4	460.62	____		___

III. Procedure:

Safety precaution: All manipulations are to be performed in the hood. There are no known hazards associated with 2,3,4,5-tetraphenylcyclopentadienone, triethylene glycol, or 1,3-diphenylacetone. Styrene (stabilized with 4-t-butyl catechol) is flammable, an irritant to the eyes and skin, and poses an inhalation hazard. Styrene is classified as a Group 2B substance, possibly carcinogenic to humans, and should only be handled in a ventilated hood with gloves and goggles. Ethyl acetate, hexane, and ethanol are flammable. Potassium hydroxide, benzil, may cause irritation to the skin, eyes, and is acutely toxic if swallowed. Chloroform-d and methylene chloride are toxic by inhalation, skin irritants, and cancer suspect agents. The product is not predicted to be hazardous according to related compounds found in the EPA ACToR database unless heated above 170 °C when styrene may be released.[2] Iodine is a skin and eye irritant and toxic by inhalation. Extra care should be taken to avoid spilling the oil baths or dripping the oil in the hoods. Oil baths should not be removed from the hood or taken off the hot plates. Do NOT dispose of the oil baths and notify your TA immediately if one is spilled.

1. Aldol Condensation Reaction

- Turn on the heater equipped with a heating block and set the temperature to 110 °C.
- To a 20 mL reaction vial, add 0.21 g of benzil, 0.21 g of 1,3-diphenylacetone, a small stir bar and 0.5 mL triethylene glycol.
- Place the vial on the heating block and wait for the solid to dissolve (or melt). The solution should turn bright yellow. Add 5-6 drops of freshly prepared 2 M KOH in ethanol. The mixture should quickly turn purple and solidify; if not add 2 more drops of 2 M KOH. Continue heating for 3-5 minutes.
- Dip a pipet (without rubber pipet bulb) into the mixture to withdraw a small amount of sample. Place the tip of the pipet into a small sample vial and run a small amount of ethyl acetate down the pipet from the bulb end to dissolve the reaction mixture. Label as R1 (Reaction1). Spot R1 against the two starting materials (benzil and 1, 3-diphenylacetone) using 10% ethyl acetate/90% hexane as the elution solvent. Observe the TLC plate under a UV lamp. Identify the aldol product on your TLC plate. If the reaction has gone to completion continue to the next step. Otherwise, consult your TA.

2. Diels-Alder Reaction

- Add 0.40 mL of styrene to the reaction mixture and continue to heat for an additional 30 minutes at 110 °C. As the Diels-Alder reaction proceeds tetraphenylcyclopentadienone will be consumed and the dark purple color should change to a tan-like color. Efficient mixing will accelerate the reaction.
- Analyze the reaction by TLC to determine if starting material is still present. Dip the tip of a pipet into the reaction mixture, dissolve in ethyl acetate in a small vial and label R2 (Reaction2). Spot on a TLC plate along with R1 and elute using 10 % ethyl acetate/90% hexane as the elution solvent. If tetraphenyl-cyclopentadienone still remains, add 1 equivalent of styrene and heat for an additional 20 minutes.

3. Work up

- Cool the mixture to room temperature and add 5 mL of ethanol to the reaction mixture. Stir with a spatula.
- Cool the mixture in an ice bath and stir, the product will precipitate from the solution.
- Collect the product using a suction filtration with a Hirsch funnel. Use a metal spatula to break apart any chunks that may have formed and spread the product well to dry. Wash the solid well with ethanol to remove any remaining styrene from the product.

4. Recrystallization

- Place the crude product and a stir bar in a 50 mL conical flask. Add 10 mL of ethanol (non-dissolving solvent) and 10 mL of ethyl acetate (dissolving solvent) In another conical flask, add about 20 ml of ethyl acetate and 2-3 boiling chips.
- Heat both flasks to gentle boiling on a hot plate. If the product is not fully soluble, add ethyl acetate drop wise until the solids are dissolved. Note: use minimum amount of ethyl acetate. Once a clear solution is observed cool to room temperature for 15-20 minutes and then in an ice bath.
- Collect the crystals by suction filtration, rinse with a minimal amount of ethanol, and dry by spreading out on the filter paper while continuing to pull the vacuum. If crystals were not formed the first laboratory day, place the conical flask inside a beaker and cover the beaker with a watch glass. Place the crystallization system inside your fumehood until the next lab session. During the next lab period, collect the crystals that have formed by suction filtration, wash with minimal amounts of ethanol and dry the crystals.

5. Analysis

- Record a melting point for the solid starting the Mel-Temp instrument at 150 °C and heating to 200 °C. Be sure to watch the process the entire time. Record any unusual observations.
- Obtain an IR spectrum
- Submit a sample for ^{1}H NMR analysis by weighing 10-20 mg of the product in a vial and dissolve with $CDCl_3$. Note, a small amount of heating may be required to dissolve the entire sample. Use the data collected to propose which product, **(3A), (3B),** or **(4)** was formed.
- An X-ray crystal structure of the major product will then be given for your interpretation.

Name: _____

Date of the experiment: _____

Datasheet for One pot Aldol Condensation and Diels-Alder Reaction

Instruction: Take photos of the measurements. After completing the experiment, show the photos to your teaching assistant and ask them to sign your printed datasheet (hardcopy, not softcopy). Submit the signed datasheet along with your lab report

Starting materials

- Actual mass of benzil used (g)_____
- Actual mass of 1,3-diphenylacetone used (g)_____
- Amount of styrene used (mL)_____

Product

- Mass of empty plastic bag (g)_____
- Mass of plastic bag + product (g) _____
- Mass of your final product (g) _____
- Percentage yield (%): _____

Observation during the experiment:

Draw your TLC plate here and label all the distances (in cm)

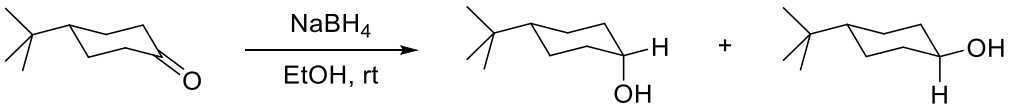
Experiment 3: Stereoselective Reduction of 4-tert-Butylcyclohexanone

I. INTRODUCTION

Reduction of carbonyl compounds is one of the key methods for preparation of alcohols, which are powerful synthetic intermediates in organic chemistry. A variety of reagents have been developed for effective and selective reductions and one of the most common reagents used in laboratories is sodium borohydride.

In general, sodium borohydride is able to reduce ketones and aldehydes with good chemoselectivity over other carbonyl compounds. In the reduction of 4-tert-butylcyclohexanone, the stereoselectivity of sodium borohydride is studied by interpretation of the ^{1}H-NMR spectrum of the product.

Reaction scheme



II. Reagents and Product Table

Reagent	MW	mmol	Density (g/mL)	amount
4-*tert*-butylcyclohexanone	154.25	___	NA	0.5 g
sodium borohydride	37.83	___	NA	0.4 g
Ethanol	46.07	NA		5 mL
4-*tert*-butylcyclohexanol	156.27	___		___

III. PROCEDURE

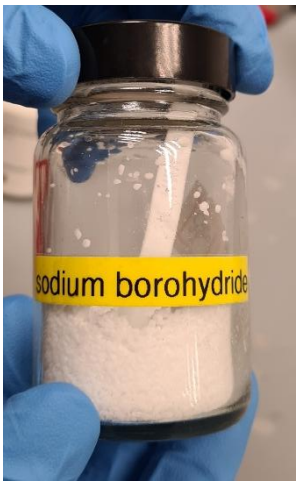
Safety Precautions: sodium borohydride, chloroform-d, ethyl acetate, hexane, diethyl ether and hydrochloric acid are all toxic and/or harmful. Ethanol, ethyl acetate, hexane and diethyl ether are flammable. Hydrochloric acid and sodium borohydride are corrosive. Sodium

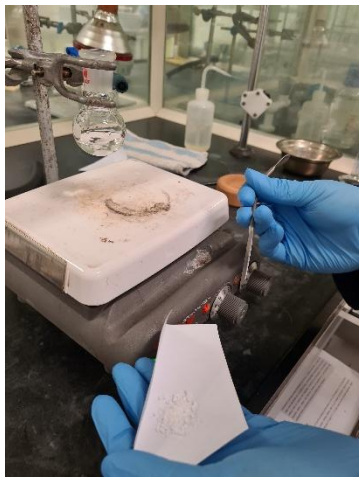
borohydride, in contact with water, emits flammable gas. Anisaldehyde stain, containing ethanol, anisaldehyde, acetic acid and sulfuric acid, is corrosive and flammable. Chloroform-d is carcinogenic. Avoid inhalation, ingestion and/or contact with skin/eyes. Perform all operations in a well-ventilated fume cupboard. Refer to the Risk Assessment for detailed safety information.

3.1 Reaction setup

- Dissolve 0.5 g of 4-*tert*-butylcyclohexanone in 5 mL of 95% ethanol in a 25 mL round bottom flask with proper stirring.

- Slowly add 0.4 g of sodium borohydride. Keep the reaction temperature between 25 and 35 ^{o}C. If it feels warm to touch, cool the mixture in a water bath (not ice-bath).

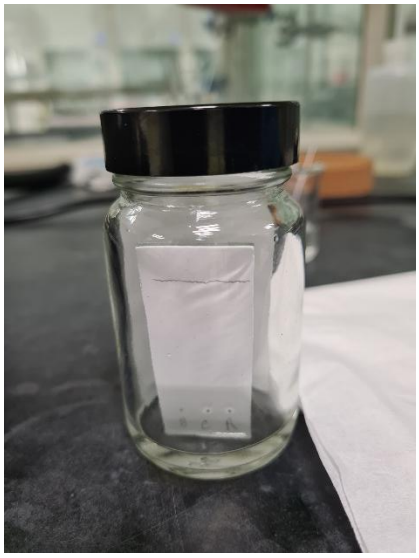




- Monitor the reaction progress by TLC using hexane – ethyl acetate (4:1). Visualize the TLC plate under a UV lamp, circle all visible spots with a pencil.

- Dip the same TLC plate in anisaldehyde stain solution, remove and heat on a hot plate until the color spots are visible. This TLC stain can provide additional information about the mixture components.

- The reaction is typically complete within one hour.



3.2 Work-up

- Once the reaction is complete, quench the reaction by careful addition of 3 mL of water.

- Add 1 mL of 3M hydrochloric acid dropwise with caution. Caution: Hydrogen gas will be generated. Gas evolution may be vigorous and should be controlled. Hydrogen gas is flammable.

- After about 10 minutes, when the evolution of hydrogen gas becomes very slow or ceases, heat the mixture on a water bath to boiling and allow the mixture to cool to room temperature. A white inorganic solid may form, which can be discarded before the extraction.

- Extract the reaction mixture with ethyl acetate (10 mL × 2).

- Extract the combined organic layers with water (10 mL × 2) then brine (10 mL).

- Dry the combined organic layer with anhydrous sodium sulfate.

- Filter the solution into a pre-weighed 100 mL rbf.

- Remove the solvent using a rotary evaporator.

- Weigh your product and transfer it to a plastic bag for submission.

3.3 Characterization

- Record the IR spectrum of the product.

- Submit a sample for ^{1}H NMR analysis of your final product.

To prepare a ^{1}H NMR sample, typically 10- 20 mg of sample is dissolved in 0.6 mL of NMR solvent ($CDCl_3$ for current experiment) in a clean and dry test tube. After the sample is dissolved, transfer the solution with a clean and dry glass dropper into a clean and dry NMR tube carefully. Cap the NMR tube and label the tube properly. Ensure the height of the solution is 4~6 cm in the NMR tube; top up if needed.

Appendix: Anisaldehyde stain (prepared by lab staff)

To 135 mL of absolute ethanol was added 5 mL of concentrated sulfuric acid, 1.5 mL of glacial acetic acid. Allow the solution to cool to room temperature. Add 3.7 mL of *p*-anisaldehyde. The solution is then stirred vigorously to ensure homogeneity. The resulting staining solution is ideally stored refrigerated in a 100 mL wide mouth jar covered with aluminum foil. Keep away from direct sunlight.

Name: _____

Lab Group_____ Fumehood _____ Date of the experiment: _____

Stereoselective Reduction of 4-tert-Butylcyclohexanone

Instruction: Take photos of the measurements. After completing the experiment, show the photos to your teaching assistant and ask them to sign your printed datasheet (hardcopy, not softcopy). Submit the signed datasheet along with your lab report

Starting materials

Mass of 4-*tert*-butylcyclohexanone added (g) _____

Mass of sodium borohydride added (g) _____

Product

Mass of the empty rbf (g)_____

Mass of product + rbf (g) _____

Mass of your final product (g) _____

Observations

Draw your TLC plate here and label all distances (in cm)

Experiment 4: Synthesis of (±) Linalool using Grignard Reaction

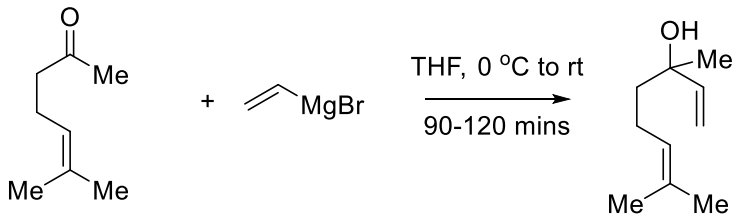
I. Introduction

Victor Grignard was awarded the Nobel Prize in Chemistry in 1912 for his discovery of the Grignard reagent. Grignard reactions have been widely used in organic synthesis due to their simplicity in carrying out the reactions experimentally and broad applications.

A Grignard reagent is represented by a general formula R-MgX, where R is an alkyl or aryl group and X is a halogen (usually chlorine, bromine, or iodine). The C–Mg bond in a Grignard reagent is not purely covalent, but it is highly polarized, making the carbon center highly nucleophilic. On the other hand, the polarization in a carbonyl group (C=O) makes the carbon center highly electrophilic. Thus, one of the many applications of a Grignard reagent is its reactions with carbonyl functions, *e.g.*, carbon dioxide, aldehydes or ketones, and esters, to afford carboxylic acids, secondary and tertiary alcohols, respectively.

Linalool is a tertiary alcohol (monoterpene natural product) that is found in the essential oils of more than 200 plant species around the globe, essentially in the genera Lavandula (Lamiaceae), Coriandrum (Apiaceae), and Cinnamomum (Lauraceae). (S)-linalool is a major constituent of the essential oils of coriander, where the (R)-linalool is present in lavender essential oil. In this experiment, you will synthesize racemic linalool from 6-methyl-5-hepten-2-one and vinyl magnesium bromide. The reaction is carried out under a nitrogen atmosphere and anhydrous conditions.

II. Reaction Scheme



1 **2** **3** (linalool)

Reagents and Product Table

Reagent	MW	mmol	density (g/mL)	amount
6-methyl-5-hepten-2-one	126.20	3.0	0.855	0.44 mL
vinyl magnesium bromide 1M solution in THF	131.25	7.5		7.5 mL
THF	72.11	NA		5 mL
Product	154.25	___		___

III. Procedure

Safety Precautions: 6-methyl-5-hepten-2-one is a flammable liquid and can cause eye irritation. Vinyl magnesium bromide is toxic, cause skin irritation and serious eye damage; in contact with water emit flammable gases. Linalool can cause skin and eye irritation. Silica gel is toxic when inhaled. Ethyl acetate, tetrahydrofuran and diethyl ether are highly flammable. Hexane is flammable and neurotoxin. Do not inhale chemicals or allow them to contact the skin. Perform all operations in a well-ventilated fume cupboard. Refer to the MSDS of chemicals used for detailed safety information

1. Reaction setup

The hot air gun provides temperature of 300- 500 ^{o}C, check and ensure there are NO FLAMMABLE CHEMICALS OR SOLVENTS inside the fumehood during the heating process



Figure 1: a Schlenk line in operation

Flame-drying the reaction apparatus

- Fit the side-neck of a 50 mL two-necked rbf equipped with a magnetic stirrer with a rubber septum. Connect the vertical neck to a stopcock adapter of a Schlenk line.

- Turn the valve on the two-way manifold of the Schlenk line to vacuum. Using a hot air gun, heat up the reaction flask to drive out any residual moisture from the reaction system.

- Let the system cool down for a few minutes then turn the valve on the two-way manifold to nitrogen. The flask is now dried and filled with inert atmosphere.

- Repeat one more time: Turn the valve on the two-way manifold of the Schlenk line to vacuum. Using a hot air gun, heat up the reaction flask to drive out any residual moisture from the reaction system. Let the system cool down for a few minutes then turn the valve on the two-way manifold to nitrogen.

- Allow the flask to cool to room temperature before adding any chemicals.



Figure 2: Flame-drying the rbf

Reaction set up: Please be CAREFUL when using needles

- Add 0.44 mL of 6-methyl-5-hepten-2-one (compound **1**, 3 mmol) and 5 mL of anhydrous THF to the round bottom flask. Then, cool the solution in an ice-water bath for about 5-10 minutes



Figure 3: Transferring THF solvent using needle

The bottle containing vinyl magnesium bromide (1M solution in THF) is packed under nitrogen by the manufacturer. The entire operation involving this reagent must be carried out under nitrogen atmosphere.

- Clamp the bottle with a retort stand and place it under nitrogen (using nitrogen ballon or a Schlenk line)
- Purge a 20 mL syringe with a long needle with nitrogen.
- Using the syringe, measure 7.5 mL of the Grignard solution. Before withdrawing the needle from the bottle, draw some nitrogen into the syringe (this will protect the solution and prevent the solution from dripping out of the needle). Add the vinyl magnesium bromide dropwise to the cooled reaction flask.

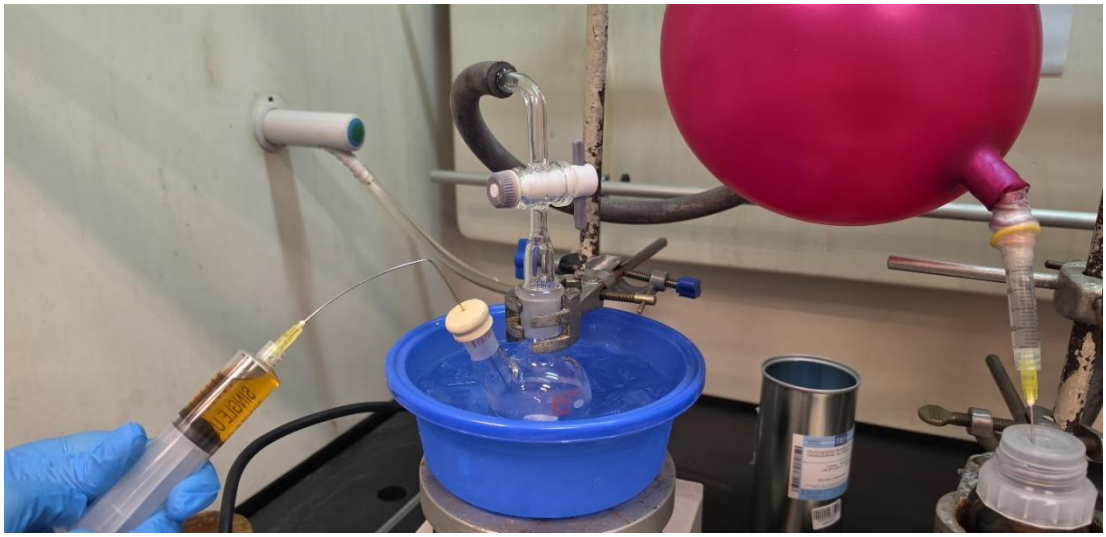
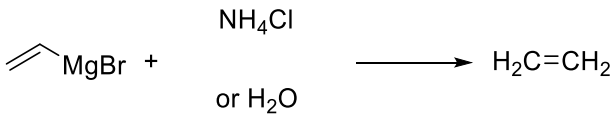


Figure 4: Adding Grignard reagent to the reaction mixture

- Once the addition is completed, remove the ice-water bath and allow the reaction mixture to stir at room temperature for 90 min.
- Quench any residual Grignard reagent in the syringe/needle with saturated ammonium chloride. The excess Grignard reagent will react with NH_4Cl or water to form ethylene gas. Perform this step in the waste fumehood. Dispose of the used syringe in a "sharp container bin".



2 Reaction workup

- Turn the valve on the two-way manifold of the Schlenk line to OFF position and disconnect the reaction flask from the Schlenk line. Use rubber septa to close the reaction flask and bring it back to your fumehood. The rest of the operation can be carried out at your usual fumehood.

- Quenching: Cool the round bottom flask in an ice-water bath and add 10 mL of saturated NH_4Cl solution to the mixture dropwise.

- Extraction: Add 10 mL of water to the round bottom flask and transfer the reaction mixture into a 150 mL separatory funnel. Rinse the round bottom flask with 20 mL ethyl acetate and transfer the rinsing to the separatory funnel.

- Shake and separate the 2 layers. Collect the aqueous layer and organic layers in separated conical flasks.

- Return the aqueous layer to the separatory funnel and extract 2 more times with 20 mL of ethyl acetate each (in total, 3 extractions with ethyl acetate). Combine the ethyl acetate layers (about 60 mL).

- Return the combined organic layer to the separatory funnel and extract with 15 mL of brine (saturated NaCl). Separate the 2 layers and place the organic layer into a dry 250 mL conical flask

- Add anhydrous Na_2SO_4 to dry the organic solvent. Let the solution stand for 15 minutes.

- TLC analysis: on the TLC plate, place 2 spots: the starting material (6-methyl-5-hepten-2-one) and the reaction mixture (using the solution that is drying in anhydrous Na_2SO_4). Perform the TLC analysis using Hexane: Ethyl Acetate 9:1 as the eluent. Stain the plate with $KMnO_4$ solution to visualise the spots. The result will be used for column chromatography (day 2).

- Filter and collect the solution in a dry pre-weighed 250 mL round bottom flask. Remove the solvent using the rotary evaporator to get the crude linalool product.

Week 2: Column chromatography

1. Preparation:

- Column eluent: In a 250mL beaker, prepare 200 mL of hexane: ethyl acetate 9:1. Obtain 2 clean Pasteur pipettes (1 for sample and the other for column eluent)

- Silica gel (Careful: silica powder is TOXIC if inhaled): in a fume cupboard, add about 20 - 30 mL of the column eluent and about 4 spoons of silica gel to a 250 mL beaker or conical flask, mix well to prepare silica gel slurry

- Column: add cotton and a layer of sand. Set up the column with a retort stand inside a fume hood. Use a 100 mL conical flask to receive the solution. Add 10 mL of the eluent to the column, open the stopcock to check the flow.

2. Packing the column:

- Pour the silica gel slurry to the column via a solid funnel. Use the column eluent to rinse and let the silica gel settle down. The length of the silica column, after it has been settled, should

be about 12-15 cm length of the column. Whenever necessary, an air pump can be used to increase the flow rate. Drain the solvent till the solvent is just above the silica gel.

- add a layer of sand (about 0.5 cm) on top of the silica gel and rinse the column. Drain the solvent till the solvent is just above the silica gel.



Figure 5: a silica gel column

3. Loading the sample:

- With a long Pasteur pipette, transfer the crude reaction mixture into the column. Open the stopcock and let the mixture absorb into the silica gel.

- Rinsing: add about 1 mL column eluent (or DCM if the solubility is low) to the round bottom flask; swirl thoroughly and transfer the liquid into the column. Open the stopcock and let the mixture absorb into the silica gel.

- Rinsing 2: Add about 2-3 mL of the column eluent the round bottom flask; swirl thoroughly and transfer the liquid into the column. Open the stopcock and let the mixture absorb into the silica gel.

4. Running flash column chromatography:

- Add about 200 ml of the prepared eluent to the column (gently – do not disturb the silica gel!). Apply air pressure and collect the eluent in test tubes.

Perform TLC to analyze the composition of each test tubes (4-6 spots can be put in a TLC plate; the capillary tube can be washed with acetone and re-used).

Because both the starting material and product are not UV active, you will need to use $KMnO_4$ stain to visualise the spots.



Figure 6: Thin Layer Chromatography Analysis

- If the product has not been eluted out, add more eluent or increase the polarity of the eluent. Continue to run the column until the entire desired product has been eluted.

- Collect all fractions containing the desired product in a dry, pre-weighed 100 mL or 250 mL round bottom flask and remove the solvent using a rotary evaporator. (you may do rotavap in batches)

- Dry the product under vacuum for another 20 minutes to remove all the solvent. Record the purified mass of the product obtained.

6. Characterization

- Calculate the amount of product obtained.
- Obtain an IR spectrum of the product.
- Submit a sample for ^1H NMR analysis.

Name: _____

Lab Group_____ Fumehood _____ Date of the experiment: _____

Datasheet for Synthesis of (±) Linalool using Grignard Reaction

Instruction: Take photos of the measurements. After completing the experiment, show the photos to your teaching assistant and ask them to sign your printed datasheet (hardcopy, not softcopy). Submit the signed datasheet along with your lab report

Starting materials:

- Amount of 6-methyl-5-hepten-2-one used (mL)_____
- Amount of vinyl magnesium bromide used (mL)_____

Crude product

- Mass of the empty rbf (g)_____
- Mass of product + rbf (g) _____
- Mass of your final product (g) _____

Purified product

- Mass of the empty rbf (g)_____
- Mass of product + rbf (g) _____
- Mass of your final product (g) _____

Observations

Draw your TLC plate here and label all distances (in cm)